Continuity

F18CD Week 8



Introduction

Introduction

Let D be an open interval and let $f : D \to \mathbb{R}$ be a function.

Then f is said to be continuous at $x = a \in \mathbb{R}$, if all below are satisfied:

1. $f(a)$ is defined, i.e., $a \in D$.
2. $\lim\limits_{x \to a} f(x)$ exists.
3. $\lim\limits_{x \to a} f(x) = f(a)$, i.e., 1. = 2.

Remark: Some author will only write 3.

One-sided continuity (at end-points)

Let $a, b \in \mathbb{R}$ with $a < b$. A function $f : [a, b] \to \mathbb{R}$ is continuous at an end-point $x = a$, or b, if

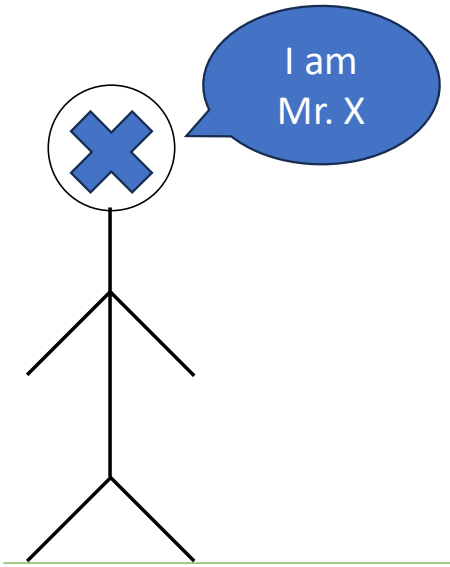
$$\lim_{x \to a^+} f(x) = f(a)\,, \text{ or } \lim_{x \to b^-} f(x) = f(b)\,.$$

If f is continuous at every point $c \in [a, b]$, then it is continuous on $[a, b]$.

The relationship of differentiability, continuity and limit

Let D be an open interval and let $f : D \to \mathbb{R}$ be a function.

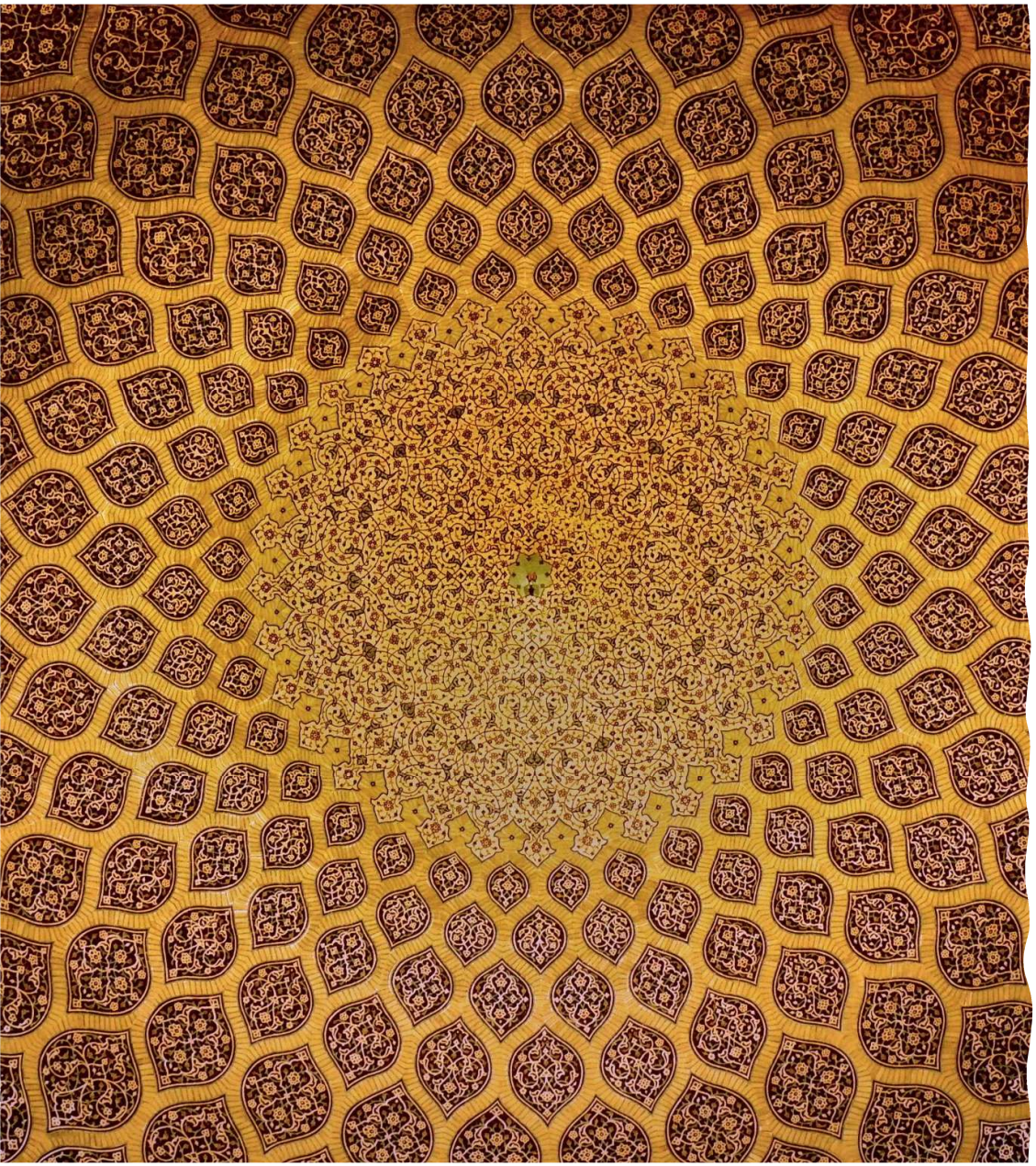
Let $x = a \in D$ *(limit points will not be considered).*



f is differentiable at $x = a$.

f is continuous at $x = a$.

Limit f exists as $x \to a$.



Definition of continuity at a point

Definition 3.3: Epsilon-delta definition of continuity at a point

Let D be an open interval and let $f : D \to \mathbb{R}$ be a function and $a \in D$. Then f is continuous at $x = a \in D$, if

$$\forall \varepsilon > 0 \;\; \exists \delta > 0 \;\ni \left|x - a\right| < \delta \Rightarrow \left|f(x) - f(a)\right| < \varepsilon \,.$$

This is the detailed complete undisputed definition. I call it the DCU.
Compare this definition to the limit's definition at a point. What are the differences?

VERY IMPORTANT YOU MUST REMEMBER

- The epsilon-delta definition MUST be used in proving a function is continuous at a point.
- Other than that, we will commonly, normally, mostly, happily use

$$\lim_{x \to a} f(x) = f(a).$$

- This is the *equivalent definition* to the epsilon-delta definition (proved in Question 9, Worksheet 5)
- We shall call this as "*the most commonly used* (MCU)"

Example 3.4: Use epsilon-delta definition (DCU)

Let $f : \mathbb{R} \to \mathbb{R}$ be a function given by $f(x) = 10x$.

Prove that f is continuous at $x = 2$.

Proof. Let $\varepsilon > 0$. Choose $\delta = \frac{\varepsilon}{10}$. Then $\left|x - 2\right| < \delta$

$$\Rightarrow \left|f(x) - f(2)\right| = \left|10x - 20\right| = 10\left|x - 2\right| < 10\delta = \varepsilon .$$

Example 3.5: Use epsilon-delta definition (DCU)

Let $f : \mathbb{R} \to \mathbb{R}$ be a function given by $f(x) = x^2$.

Prove that f is continuous at $x = 3$.

Proof. Let $\varepsilon > 0$. Choose $\delta = \min\{1, \frac{\varepsilon}{7}\}$. Then $\left|x - 3\right| < \delta \Rightarrow$

$$\left|f(x) - f(3)\right| = \left|x^2 - 9\right| = \left|x - 3\right|\left|x + 3\right| \leq \left|x - 3\right|\Big(\left|x - 3\right| + 6\Big) < \varepsilon\,.$$

Example: Piecewise function (DCU)

Let $f : \mathbb{R} \to \mathbb{R}$ be a function given by

$$f(x) = \begin{cases} \frac{2x^2}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}.$$

Prove that f is continuous at $x = 0$.

Proof. If $x = 0$, then $\left|f(x) - f(0)\right| = 0 < \varepsilon$ for all $\varepsilon > 0$ whenever $\left|x - 0\right| = 0 < \delta$.

Note that this case is trivial and we normally omit it.

If $x \neq 0$. then set $\delta = \frac{\varepsilon}{2}$, where $\varepsilon > 0$ is arbitrary. Then

$$0 < \left|x\right| < \delta \Rightarrow \left|f(x) - f(0)\right| = \left|2x\right| = 2\left|x\right| < 2\delta = \varepsilon .$$

Note that we take $f(x) = \frac{2x^2}{x} = 2x, x \neq 0$ here.

Definition: Epsilon-delta definition of continuity at end points

Let $f : \left[a, b\right] \to \mathbb{R}$ be a function.

Then f is continuous at end points $x = a$ if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ni a \leq x < a + \delta \Rightarrow \left| f(x) - f(a) \right| < \varepsilon \, .$$

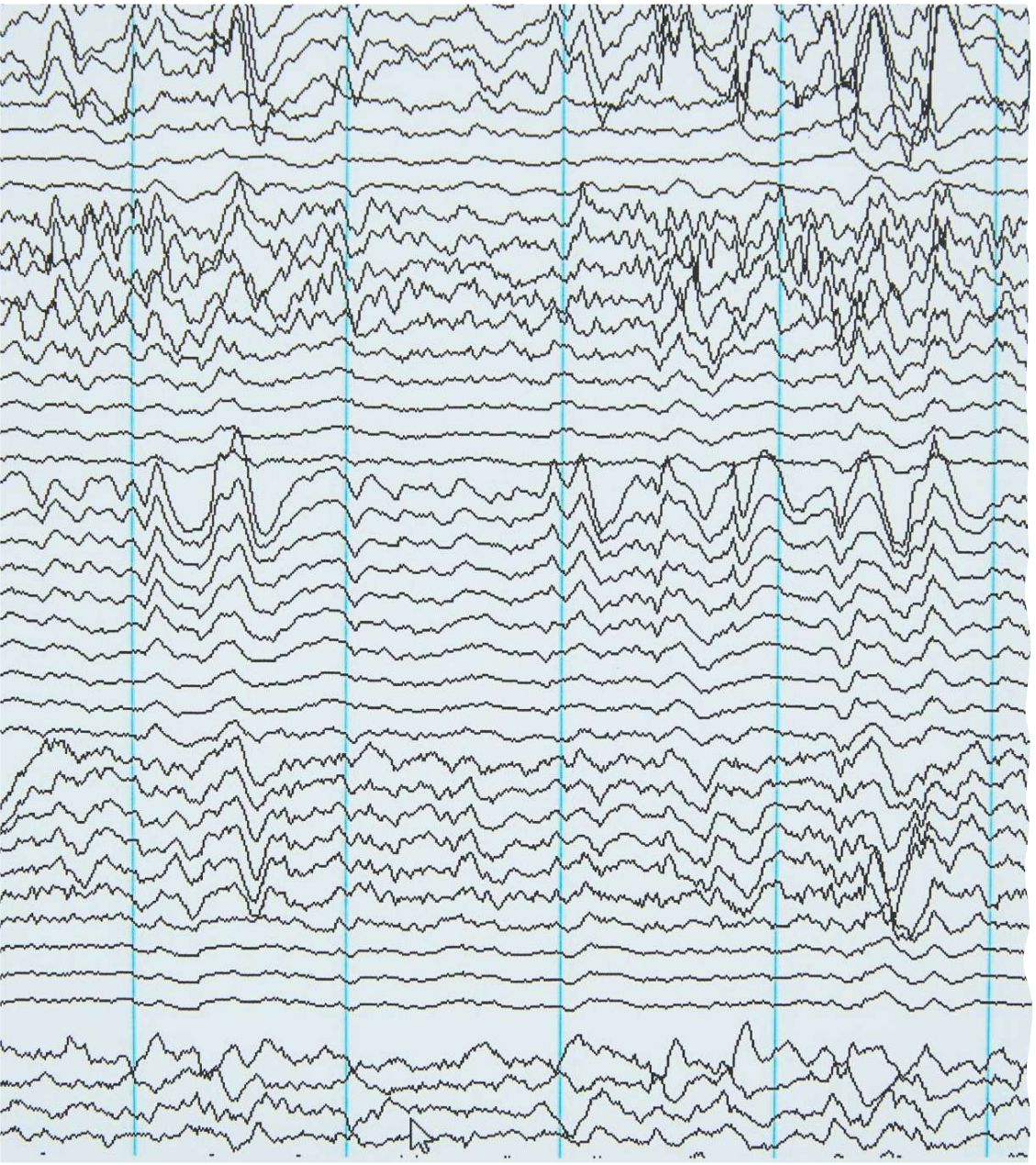
Similarly, f is continuous at end point $x = b$ if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ni b - \delta < x \leq b \Rightarrow \left| f(x) - f(b) \right| < \varepsilon \, .$$

Exercise (MCU)

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined $f(x) = \begin{cases} x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$.

Prove that f is continuous at $x = 0$.



Continuous on an interval

Definition 3.6: Continuous on interval

Let $f : D \to \mathbb{R}$ be a function.

Then f is said to be continuous (everywhere) on $I \subset D$ if f is continuous at each point $c \in I$.

Examples

Functions like $\sin x$, $\cos x$, e^x are continuous everywhere on $\mathbb{R}$.

Functions like $\ln x$, $\tan x$ are continuous but not everywhere on $\mathbb{R}$,

but $\ln x$ is continuous everywhere on $\mathbb{R}^+$, and

$\tan x$ is continuous everywhere on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Continuous is like one stroke (Mandarin: 一笔画) (Malay: *selanjar*).

Example: Use the MCU

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} 0, & x < -1 \\ ax^2 + bx, & -1 \leq x \leq 1 \\ cx, & x > 1 \end{cases},$$

where a, b, c are constants.

Find relationship of a, b, c if f is continuous everywhere on $\mathbb{R}$.

Solution. Consider $x = \pm 1$. (Why?) To have f is continuous everywhere on $\mathbb{R}$,

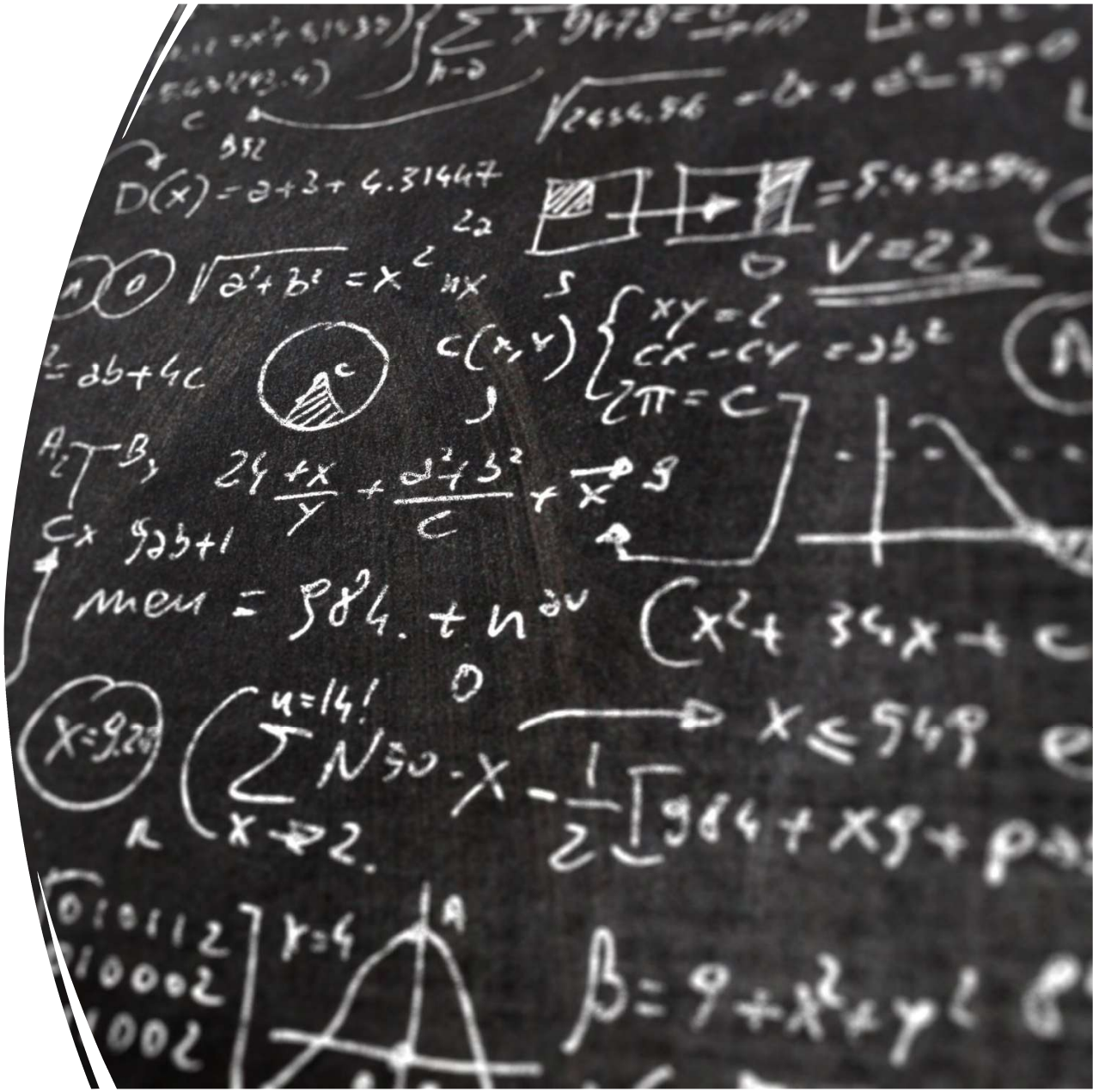
$$\lim_{x \to -1} f(x) = f(-1) \text{ and } \lim_{x \to 1} f(x) = f(1).$$

So, $\lim\limits_{x \to -1^-} f(x) = \lim\limits_{x \to -1^-} 0 = 0 = a - b = \lim\limits_{x \to -1^+} (ax^2 + bx) = \lim\limits_{x \to -1^+} f(x) \Rightarrow a = b$ (1)

and $\lim\limits_{x \to 1^-} f(x) = \lim\limits_{x \to 1^-} (ax^2 + bx) = a + b = c = \lim\limits_{x \to 1^+} (cx) = \lim\limits_{x \to 1^+} f(x) \Rightarrow a + b = c$ (2)

Hence, f is continuous everywhere on $\mathbb{R}$ if (1) and (2) are satisfied, i.e., $c = 2a = 2b$.

Properties of continuous functions



Theorem 4.1: Properties of continuous functions

Let $f, g : D \to \mathbb{R}$ be continuous functions at the point $c \in D$.

Then the following functions are all continuous at the point c.

(i) $f + g$ (ii) fg (iii) $1/g$, proving $g(c) \neq 0$

(iv) f/g, proving $g(c) \neq 0$ (v) λf , $\lambda \in \mathbb{R}$

Proof. (i) Given that f and g are continuous at c, i.e.,

$$\forall \varepsilon_1 > 0 \ \exists \delta_1 > 0 \ni \left|x - c\right| < \delta_1 \Rightarrow \left|f(x) - f(c)\right| < \varepsilon_1$$
$$\forall \varepsilon_2 > 0 \ \exists \delta_2 > 0 \ni \left|x - c\right| < \delta_2 \Rightarrow \left|g(x) - g(c)\right| < \varepsilon_2$$

Choose $\varepsilon_1 = \varepsilon_2 = \frac{\varepsilon}{2}$ and $\delta = \min\left\{\delta_1, \delta_2\right\}$. Then

$$\left|f(x) + g(x) - \left(f(c) + g(c)\right)\right| \leq \left|f(x) - f(c)\right| + \left|g(x) - g(c)\right| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

whenever $\left|x - c\right| < \delta$.

For (ii), use the following fact:

$$\boxed{\boxed{\Big|\big|a\big| - \big|b\big|\Big| \leq \big|a - b\big| \Leftrightarrow -\big|a - b\big| \leq \big|a\big| - \big|b\big| \leq \big|a - b\big|}}$$

Proof. (ii) Given that f and g are continuous at c, i.e.,

$$\forall \varepsilon_1 > 0 \ \exists \delta_1 > 0 \ni \left|x-c\right| < \delta_1 \Rightarrow \left|f(x)-f(c)\right| < \varepsilon_1$$
$$\forall \varepsilon_2 > 0 \ \exists \delta_2 > 0 \ni \left|x-c\right| < \delta_2 \Rightarrow \left|g(x)-g(c)\right| < \varepsilon_2$$

Note that $\left|x-c\right| < \delta_3 \Rightarrow \left|f(x)-f(c)\right| < 1$ (pick $\varepsilon_3 = 1$)

$$\Rightarrow \left|f(x)\right| - \left|f(c)\right| \le \left|f(x)-f(c)\right| < 1 \quad \because \left|a\right| - \left|b\right| \le \left|a-b\right|$$
$$\Rightarrow \left|f(x)\right| < \left|f(c)\right| + 1$$

Choose $\varepsilon_1 = \frac{\varepsilon}{2\left(\left|g(c)\right|+1\right)}$, $\varepsilon_2 = \frac{\varepsilon}{2\left(\left|f(c)\right|+1\right)}$ and $\delta = \min\left\{\delta_1, \delta_2, \delta_3\right\}$.

$$
\begin{aligned}
\text{Then } \Big|f(x)g(x) - f(c)g(c)\Big| & \\
&= \Big|f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c)\Big| \\
&\le \Big|f(x)\Big|\Big|g(x) - g(c)\Big| + \Big|g(c)\Big|\Big|f(x) - f(c)\Big| \\
&< \Big(\Big|f(c)\Big| + 1\Big) \times \frac{\varepsilon}{2\Big(\Big|f(c)\Big|+1\Big)} + \Big|g(c)\Big| \times \frac{\varepsilon}{2\Big(\Big|g(c)\Big|+1\Big)} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon
\end{aligned}
$$

whenever $\Big|x - c\Big| < \delta$.

(iii) Note that $g(c) \neq 0$. So $\left|x-c\right| < \delta_3$ implies that

$$\begin{aligned} \left|g(c)-g(x)\right| = \left|g(x)-g(c)\right| < \varepsilon_3 = \tfrac{1}{2}\left|g(c)\right| \neq 0 \\ \Rightarrow \left|g(c)\right| - \left|g(x)\right| < \tfrac{1}{2}\left|g(c)\right| \\ \Rightarrow \tfrac{1}{2}\left|g(c)\right| < \left|g(x)\right| \end{aligned}$$

Given that g is continuous at c, i.e.,

$$\forall \varepsilon_2 > 0 \ \exists \delta_2 > 0 \ni \left|x-c\right| < \delta_2 \Rightarrow \left|g(x)-g(c)\right| < \varepsilon_2$$

Choose $\varepsilon_2 = \tfrac{\varepsilon}{2}\left|g(c)\right|^2$ and $\delta = \min\left\{\delta_2, \delta_3\right\}$.

$$\left|\frac{1}{g(x)} - \frac{1}{g(c)}\right| = \left|\frac{g(c)-g(x)}{g(c)g(x)}\right|$$

$$= \frac{1}{\left|g(c)\right|\left|g(x)\right|}\left|g(x) - g(c)\right|$$

$$< \frac{2}{\left|g(c)\right|^2}\left(\frac{\varepsilon}{2}\left|g(c)\right|^2\right) = \varepsilon$$

whenever $\left|x - c\right| < \delta$.

(iv) This is immediate from (ii) and (iii).

(v) Given $\forall \frac{\varepsilon}{|\lambda|+1} > 0$ $\exists \delta > 0$ э

$$\left|x - c\right| < \delta \Rightarrow \left|f(x) - f(c)\right| < \frac{\varepsilon}{|\lambda|+1}$$

Then $\left|\lambda f(x) - \lambda f(c)\right| \leq \left|\lambda\right|\left|f(x) - f(c)\right| < \frac{|\lambda|\varepsilon}{|\lambda|+1} < \varepsilon$

whenever $\left|x - c\right| < \delta$.

Example 4.2: Polynomial

Recall $\lim_{x \to a} P(x) = P(a)$. Thus, polynomials are continuous at any point

Proof. Exercise

Theorem 4.3: Continuity on composite functions

Let $f : D_1 \to \mathbb{R}$ and $g : D_2 \to \mathbb{R}$ be functions with the range of f,

$$f(D_1) := \left\{ f(x) : x \in D_1 \right\} \subseteq D_2 \,.$$

Suppose f is continuous at $c \in D_1$, and g is continuous at $f(c) \in f(D_1)$.

Then $g \circ f$ is continuous at $c \in D_1$.

Proof. Given f is continuous at $x = c \in D_1$:

$\forall \varepsilon' > 0 \ \exists \delta > 0 \ni \left|x - c\right| < \delta \Rightarrow \left|f(x) - f(c)\right| < \varepsilon', x \in D_1$.

Given g is continuous at $y = f(c) \in f(D_1)$:

$\forall \varepsilon > 0 \ \exists \delta' > 0 \ni \left|y - f(c)\right| < \delta' \Rightarrow \left|g(y) - g(f(c))\right| < \varepsilon, y \in f(D_1)$.

Since $y \in f(D_1)$, $\exists x \in D_1 \ni f(x) = y$.

So, if we pick $\varepsilon' = \delta'$, we have $\forall \varepsilon > 0 \ \exists \delta > 0$ such that

$\left|x - c\right| < \delta \Rightarrow \left|f(x) - f(c)\right| < \varepsilon' = \delta' \Rightarrow \left|g(f(x)) - g(f(c))\right| < \varepsilon$.

Hence, $g \circ f$ is continuous at $c \in D_1$.

Example 4.4

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = \sin x$.

Prove that f is continuous everywhere on $\mathbb{R}$.

Proof idea

1. Let $g(x) = x - a$ and prove that $g(x)$ is continuous at a.
2. Let $f(x) = \sin x$ and prove that $f(x)$ is continuous at $g(a) = 0$.
3. Let $h(x) = \cos x$ and prove that $h(x)$ is continuous at $g(a) = 0$.
4. Theorem 4.3 guarantees $f(g(x)) = \sin(x - a)$ and $h(g(x)) = \cos(x - a)$ are continuous at a.
5. Use identity $\sin(x - a + a) \equiv \sin a \cos(x - a) + \cos a \sin(x - a)$.
6. Apply Theorem 4.1 to show that $\sin x$ continuous at a.

$|\sin\ x| \leq |x|$

From $\cos x < \frac{\sin x}{x} < 1\ \forall x \in (0, \frac{\pi}{2})$, it follows that

the result is also true $\forall x \in (-\frac{\pi}{2}, 0)$. (Why?)

Since $-1 < \cos x \leq 1$ for $\forall x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, it follows that

$$-1 < \cos x < \frac{\sin x}{x} < 1 \Rightarrow \left|\frac{\sin x}{x}\right| < 1 \Rightarrow \left|\sin x\right| \leq \left|x\right|$$

because $\left|\sin x\right| \leq \left|x\right|$ when $x = 0$.

(sufficient for delta)

$|\cos\ x - 1| \leq |x\,|$

For $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $\cos x = 1 - 2\sin^2\left(\frac{x}{2}\right)$ implies that

$$\left|\cos x - 1\right| = 1 - \cos x = 2\sin^2\left(\frac{x}{2}\right) = 2\left|\sin\left(\frac{x}{2}\right)\right|^2 \leq 2\left|\frac{x}{2}\right|^2 = \frac{x^2}{2} *.$$

From Q14, Worksheet 3. Note that for $x \in \left(-1,1\right) \subset \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

$\left|x\right| \geq \left|x\right|^2 \geq \left|x\right|^3 \geq \cdots$. Thus, $\left(\frac{x}{\sqrt{2}}\right)^2 = \left|\frac{x}{\sqrt{2}}\right|^2 \leq \left|\frac{x}{\sqrt{2}}\right| \leq \left|x\right|$.

Hence, $\left|\cos x - 1\right| \leq \left|x\right|$ for $x \in \left(-1,1\right)$.

Remark: We can also use the fact * to prove that $\lim\limits_{x\to 0} \cos x = 1$ i.e., pick $\delta = \sqrt{2\varepsilon}$.

Proof

Let $\delta = \min\{1, \varepsilon\}$ and for any $\varepsilon > 0$. Let $a \in \mathbb{R}$ be arbitrary. Show that

(a) g is continuous at a, i.e., $|x - a| < \delta \Rightarrow |g(x) - g(a)| = |x - a| < \delta = \varepsilon$.

(b) f is continuous at 0, i.e., $|x - 0| < \delta \Rightarrow |\sin x - 0| \leq |x| < \delta = \varepsilon$.

(c) h is continuous at 0, i.e., $|x - 0| < \delta \Rightarrow |\cos x - 1| \leq |x| < \delta = \varepsilon$.

Proof (cont.)

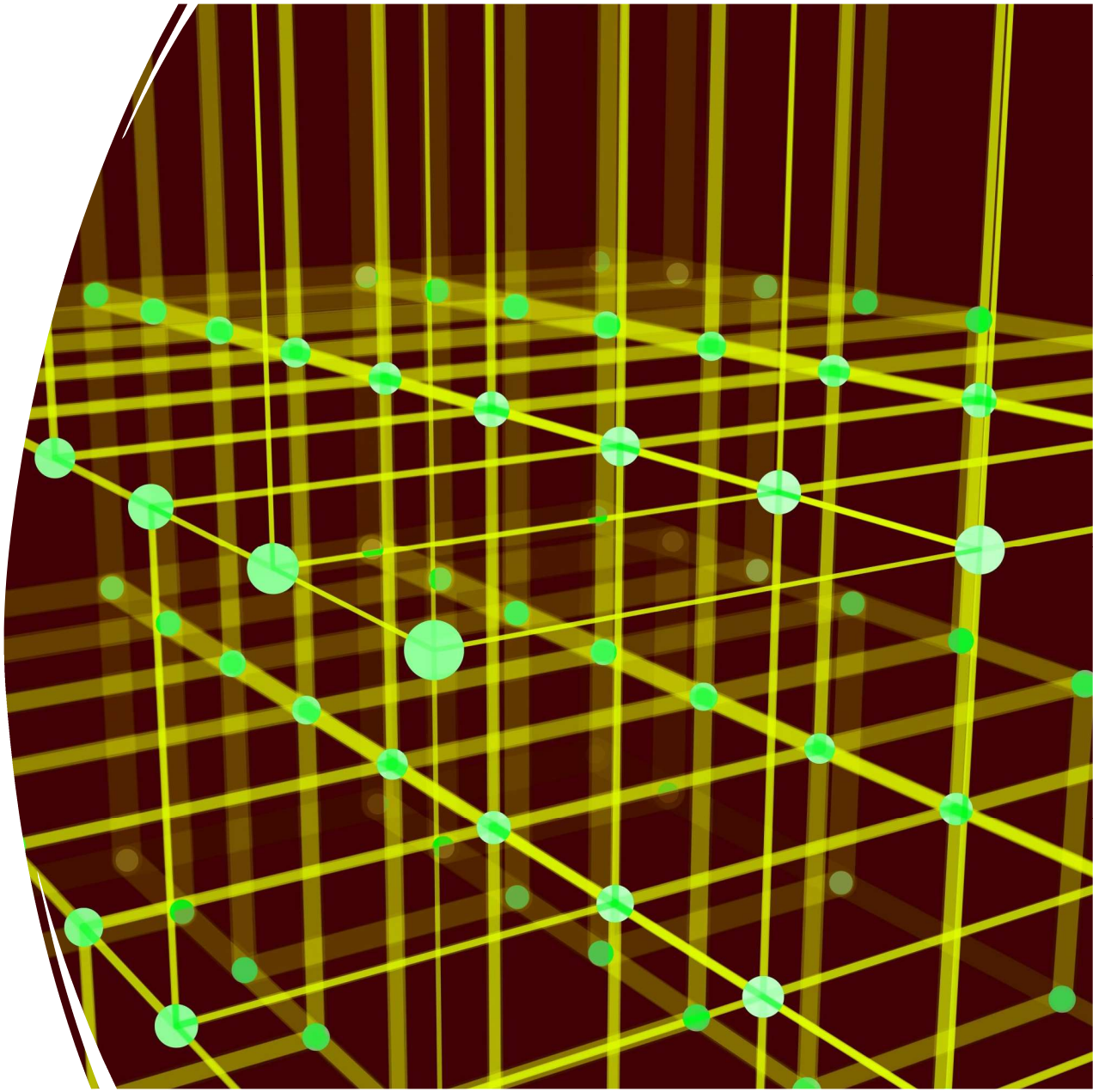
(d) By Theorem 4.3: $f \circ g = \sin(x-a)$, $h \circ g = \cos(x-a)$ are continuous at a.

(e) Use identity $\sin x \equiv \sin(x-a+a) \equiv \sin(x-a)\cos a + \sin a \cos(x-a)$.

(f) By Theorem 4.1, $f(x) = \sin x$ is continuous everywhere on $\mathbb{R}$.

Note: $\cos a$ and $\sin a$ are constants and continuous everywhere on $\mathbb{R}$
by Example 4.2 (polynomial).

Function's discontinuity at a point



Recall the MCU to define the continuity

- 1. $f\,(a)$ is defined.
- 2. limit of $f\,(x)$ when x tends to a exists.
- 3. $\lim\limits_{x \to a} f(x) = f(a)$, i.e., 1. is equal to 2.

The non-compliance to any rule above will give the function a discontinuity at the point $x = a$.

The MCU definition on discontinuity

A function is said to be discontinuous at a point $x = a$ if *either one* of the following occur:

1. $f(a)$ is undefined.

2. $\lim\limits_{x \to a} f(x) \nexists$.

3. $f(a)$ is defined and $\lim\limits_{x \to a} f(x) \exists$ but $\lim\limits_{x \to a} f(x) \neq f(a)$.

Definition 5.3: Definition of discontinuity (DCU)

A function $f : D \to \mathbb{R}$ is said to be discontinuous at a point $x = c \in \bar{D}$ if it is either

(i) not continuous at the point $c \in \bar{D}$, or

(ii) $c \notin D$.

Remarks

1. (i) is the noncompliance of limit definition, i.e.,

   if limit does not exist at c, then the function is not continuous at c.

2. (i) is the noncompliance of MCU 2 or 3; and (ii) is the noncompliance of MCU 1.

Definition 5.3: Definition of discontinuity (DCU)

Let $f : D \to \mathbb{R}$ be a function. Let $c \in \bar{D}$, $x \in D$.

Then f is not continuous at c if $c \notin D$ or $\forall \delta > 0$

$\exists \varepsilon > 0 \ni 0 < \left|x - c\right| < \delta$ but $\left|f(x) - f(c)\right| \geq \varepsilon$.

These three cases corresponding to each conditions in MCU

Let $f : D \to \mathbb{R}$ be a function. Let $c \in \bar{D}$, $x \in D$.

Case 1: If $c \notin D$, then f is not continuous at c.

Case 2: If $c \in \bar{D} \setminus D$, then f is not continuous at c if

$\forall \delta > 0 \; \exists \varepsilon > 0 \ni 0 < \left|x - c\right| < \delta$ but $\left|f(x) - f(c)\right| \geq \varepsilon$.

Case 3: If $c \in D$, then f is not continuous at c if

$\forall \delta > 0 \; \exists \varepsilon > 0 \ni \left|x - c\right| < \delta$ but $\left|f(x) - f(c)\right| \geq \varepsilon$.

We normally consolidate Case 2 and Case 3.

MCU vs DCU in discontinuity

- Unlike the case when we prove the continuity of a function at a point, we must use the DCU; here we are free to use MCU when proving discontinuity.
- But for some cases, we will demonstrate the advantage of using the DCU.

Example: Use the MCU

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined $f(x) = \begin{cases} x, & x \neq 0 \\ 1, & x = 0 \end{cases}$.

Prove that f is discontinuous at $x = 0$.

Proof. Since $\lim_{x \to 0} f(x) = 0 \neq 1 = f(0)$, it follows that f is discontinuous at $x = 0$.

Note that $\lim_{x \to 0} f(x) = 0$ exists because both left and right limit equal to 0.

Example: Use the MCU

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} 3x & x \geq 0 \\ 3 & x < 0 \end{cases}.$$

Show that f is not continuous at $x = 0$.

Proof. Consider one-sided limits:

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} 3x = 0 \neq 3 = \lim_{x \to 0^-} 3 = \lim_{x \to 0^-} f(x).$$

So $\lim_{x \to 0} f(x)$ does not exist at $x = 0$ and hence, f is not continuous at $x = 0$.

Example 5.4: Use the DCU

Let $f : \mathbb{R}^+ \to \mathbb{R}^+$ given by $f(x) = \frac{1}{x}$. Show that f is discontinuous at $x = 0$.

Proof. Then 0 is a limit point of $\mathbb{R}^+$ but $0 \notin \mathbb{R}^+$ (Definition 5.3(ii)).

Hence, f is discontinuous at the limit point $x = 0$.

Using the MCU: $f(x) = 1/x$ is not continuous at $x = 0$ because $f(0)$ is not defined.

Example 5.5. Use the MCU

Let $f:[0,1]\to\mathbb{R}$ be defined by $f(x)=\begin{cases}\sin\left(\frac{1}{x}\right), x\in(0,1]\\ 0, \quad x=0\end{cases}$.

Prove that f is discontinuous at 0.

Proof. Note that $f(0)$ is defined and we have proved that $\lim\limits_{x\to 0}\sin\left(\frac{1}{x}\right)\not\exists$ (see slide #79 in Week 7 lecture). So, f is discontinuous at 0.

Example 5.5: Use the DCU

Proof. Let $\varepsilon = \frac{1}{2}$. Then by Archimedean property,

$\forall \delta > 0 \; \exists n \in \mathbb{N}$ such that $0 < \frac{2}{(2n-1)\pi} < \delta$.

Let $x = \frac{2}{(2n-1)\pi}$. Since

$$\left|\sin\left(\frac{1}{x}\right)\right| = \left|\sin\frac{\pi(2n-1)}{2}\right| = 1 \geq \frac{1}{2},$$

it follows that f is discontinuous at 0.

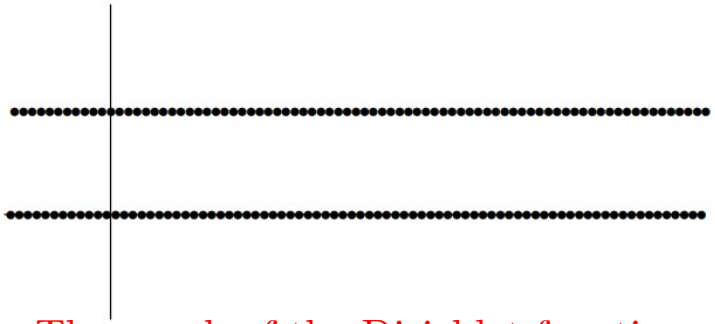
$\sin x = 0 \Leftrightarrow x = n\pi, n \in \mathbb{Z}$
$\sin x = 1 \Leftrightarrow x = \frac{4n+1}{2}\pi, n \in \mathbb{Z}$
$\sin x = -1 \Leftrightarrow x = \frac{4n-1}{2}\pi, n \in \mathbb{Z}$
$\left|\sin x\right| = 1 \Leftrightarrow \sin x = \pm 1 \Leftrightarrow x = \frac{2n-1}{2}\pi$ or $x = \frac{2n+1}{2}\pi$

Example: Use the DCU

The Dirichlet function $d : \mathbb{R} \to \mathbb{R}$

$$d(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}.$$

is nowhere continuous.



The graph of the Dirichlet function

How to pronounce Dirichlet (Johann Peter Gustav Lejeune Dirichlet - German mathematician) : The (D) Rich (Fillet)

Proof

Let $\delta = \frac{1}{2}$ and $c \in \mathbb{Q}$. Then $d(c) = 1$.

Since irrationals are dense in $\mathbb{R}$, it follows that

$\exists x \in \mathbb{R} \setminus \mathbb{Q} \ni 0 < \left|x - c\right| < \delta \ \forall \delta > 0$, and

$$\left|d(x) - d(c)\right| = \left|0 - 1\right| = 1 \geq \frac{1}{2}.$$

Proof (cont.)

Let $\delta = \frac{1}{2}$ and $c \in \mathbb{R} \setminus \mathbb{Q}$. Then $d(c) = 0$.

Since rationals are dense in $\mathbb{R}$, it follows that

$\exists x \in \mathbb{R} \ni 0 < \left|x - c\right| < \delta \ \forall \delta > 0$, and

$$\left|d(x) - d(c)\right| = \left|1 - 0\right| = 1 \geq \tfrac{1}{2}.$$

Remarks on discontinuity

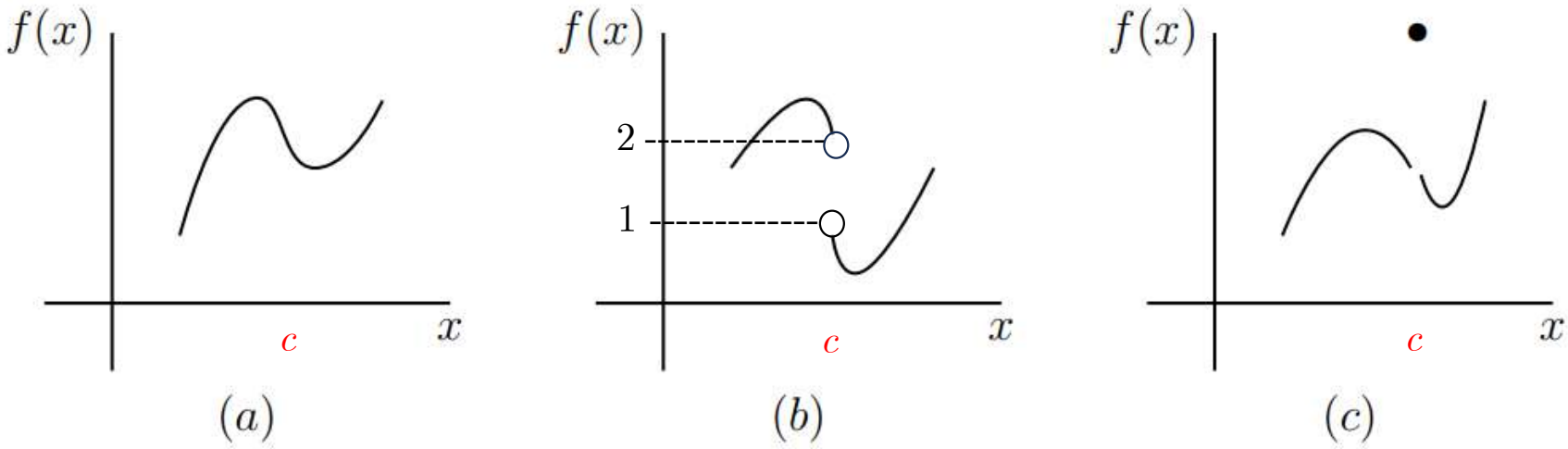
A function f can be discontinuous at a in various ways:

(a) oscillatory, so that the one-sided limits do not exist

(b) the one-sided limits may exist but not be equal to each other (so the limit does not exist at that point) — in this case f is said to have a jump discontinuity

(c) the limit exists but it is not equal to the value $f(a)$ — in this case one can modify the function to give a continuous function, if required.

Graphic illustration

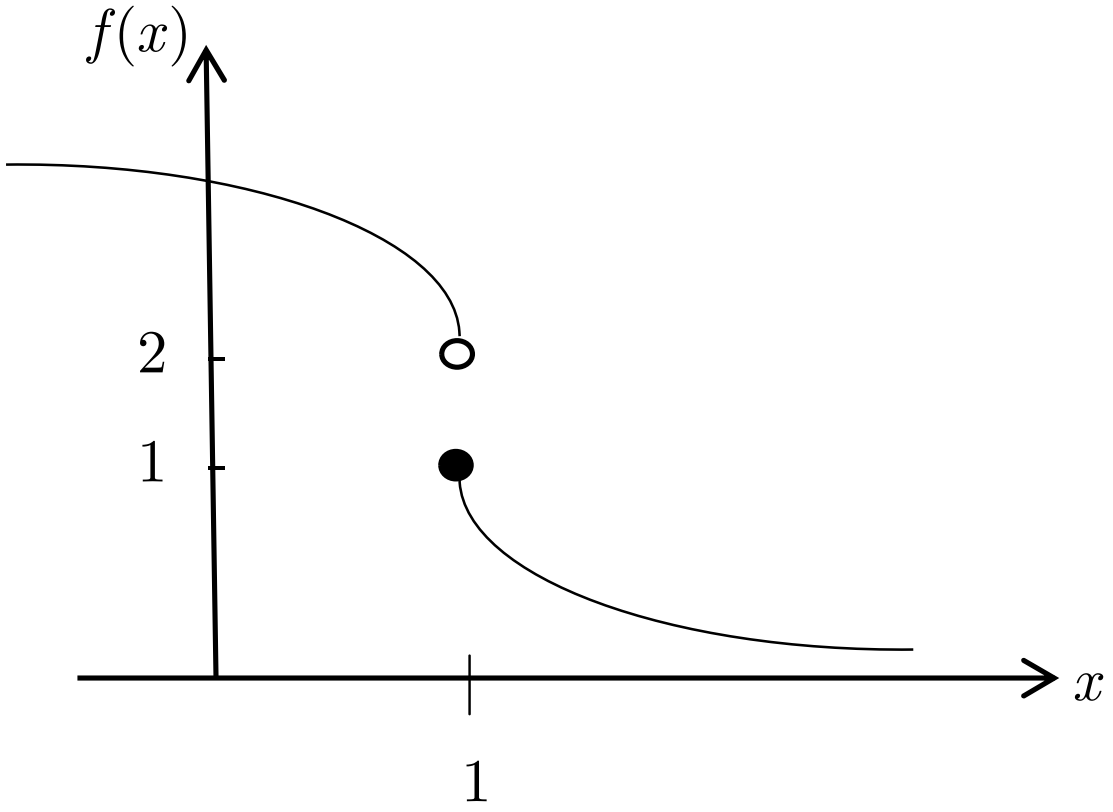


(a) Continuous at $x = c$.

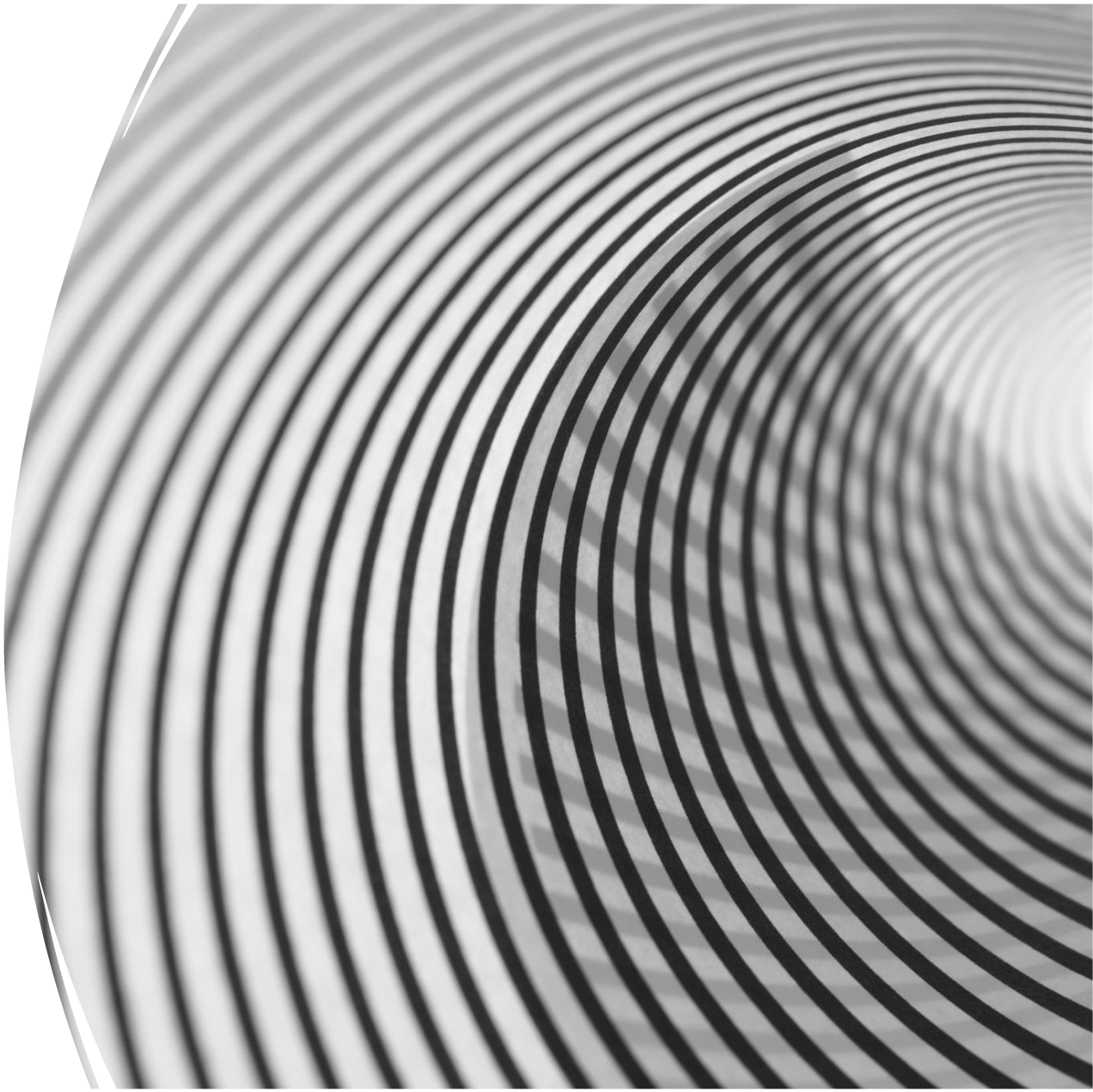
(b) Not continuous at $x = c$ due to (i) (and (ii)) in DCU (failed to comply 1. and 2. in MCU).

(c) Not continuous at $x = c$ due to (i) in DCU (or failed to comply 3. in MCU).

Exercise: Explain why $f(x)$ is discontinuous at $x = 1$



Sequential continuity



Definition: Sequential continuity

Let $D \subseteq \mathbb{R}$ be a set. Let $a : \mathbb{N} \to D$ be a convergent sequence with $\lim\left(a_n\right) = a$.

A function $f : D \to \mathbb{R}$ is said to be sequentially continuous at a if $\lim\left(f\left(a_n\right)\right) = f(a)$,

i.e., $$\boxed{\lim\left(a_n\right) = a \Rightarrow \lim\left(f\left(a_n\right)\right) = f(a)}.$$

Example: Sequentially continuous at a point

Let $a : \mathbb{N} \to \mathbb{R}$ be defined as $\left(a_n\right) = \left(\frac{1}{n}\right)$. Then $\left(a_n\right)$ converges to 0.

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = 2x$. Then $f\left(a_n\right) = \frac{2}{n}$ and $\left(\frac{2}{n}\right)$ converges to 0.

So $\lim\left(f\left(a_n\right)\right) = 0 = f(0)$ and thus f is sequentially continuous at 0.

Example: Not sequentially continuous at a point

Let $a : \mathbb{N} \to \mathbb{R}$ be defined as $\left(a_n\right) = \left(\frac{1}{n}\right)$. Then $\left(a_n\right)$ converges to 0.

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = \begin{cases} 2x, & x \neq 0 \\ 1, & x = 0 \end{cases}$.

Then $f\left(a_n\right) = \frac{2}{n}$ (because $a_n \neq 0 \ \forall n \in \mathbb{N}$) and $\left(\frac{2}{n}\right)$ converges to 0.

So, $\lim\left(f\left(a_n\right)\right) = 0 \neq 1 = f(0)$ and thus f is not sequentially continuous at 0.

Recall Week 7 slides: Theorem 7.2

Let $f : D \to \mathbb{R}$ and $c \in \overline{D}$. Let $a : \mathbb{N} \to D \setminus \{c\}$,

i.e., $a_n \in D \setminus \{c\}\ \forall n \in \mathbb{N}$. Then

$$\lim_{x \to c} f(x) = L \Leftrightarrow \lim\left(f\left(a_n\right)\right) = L\ \forall\left(a_n\right) \to c\,.$$

Proof. Let $f : D \to \mathbb{R}$ and $c \in \bar{D}$. Let $a_n \in D \setminus \{c\}\ \forall n \in \mathbb{N}$.

$\Rightarrow$) Suppose $\lim_{x \to c} f(x) = L$. Then $\forall \varepsilon > 0\ \exists \delta > 0$ such that

$$0 < \left|x - c\right| < \delta \Rightarrow \left|f(x) - L\right| < \varepsilon\,,\ x \in D\,.$$

Given that sequence $\left(a_n\right)$ converges to c. So $\forall \sigma > 0\ \exists N \in \mathbb{N}$ such that $n > N \Rightarrow 0 < \left|a_n - c\right| < \sigma\ \forall n \in \mathbb{N}$.

Now take $x = a_n$ and choose $\sigma = \delta$, then $\forall \varepsilon > 0 \ \exists \delta > 0$

$$n > N \Rightarrow 0 < \left|a_n - c\right| < \delta \Rightarrow \left|f(a_n) - L\right| < \varepsilon \ \forall n \in \mathbb{N}.$$

Thus, $\lim\left(f(a_n)\right) = L$.

$\Leftarrow$) Assume the following compound statement is true:

There exists a sequence $\left(a_n\right)$ with $\lim\left(f(a_n)\right) = L$, and $\lim\limits_{x\to c} f(x) \neq L$.

Given $\lim\left(a_n\right) = c$: $\forall\sigma > 0\ \exists M_1 \in \mathbb{N} \ni n > M_1 \Rightarrow 0 < \left|a_n - c\right| < \sigma\ \forall n \in \mathbb{N}$.

and $\lim\left(f(a_n)\right) = L$: $\forall\mu > 0\ \exists M_2 \in \mathbb{N} \ni n > M_2 \Rightarrow \left|f(a_n) - L\right| < \mu\ \forall n \in \mathbb{N}$.

$\lim\limits_{x\to c} f(x) \neq L$: $\exists\varepsilon > 0$ such that $0 < \left|x - c\right| < \delta$ and $\left|f(x) - L\right| \geq \varepsilon\ \forall\delta > 0$.

Take $x = a_n$, and pick $\delta = \sigma$, $\mu = \varepsilon$. Then a contradiction is occured.

Theorem: Characterisation of Continuity

Let $f : D \to \mathbb{R}$, and let $c \in D$. Then the function f is continuous at c if and only if any one of the following conditions is met:

(i) $\forall \varepsilon > 0 \ \exists \delta > 0 \ni \left|x - c\right| < \delta \Rightarrow \left|f(x) - f(c)\right| < \varepsilon \ \forall x \in D$

(ii) $\lim_{x \to c} f(x) = f(c)$

(iii) $\forall V_{\varepsilon}\left(f(c)\right) \exists V_{\delta}(c) \ni x \in V_{\delta}(c) \Rightarrow f(x) \in V_{\varepsilon}\left(f(c)\right) \forall x \in D$

(iv) Let $x_n \in D$. If $\left(x_n\right) \to c$, then $f\left(x_n\right) \to f\left(c\right)$.

Remark: f is continuous at $x = a \Leftrightarrow f$ is sequentially continuous at $x = a$.

Theorem: f is continuous at $x = a$
$\Leftrightarrow$ f is sequentially continuous at $x = a$.

Proof. Let $L = f(a)$ and modify the Theorem 7.2
with c now is an element of D (no longer a limit point).

Revisit: Corollary 7.3

Corollary 7.3. If there exist two distinct sequences $\left(a_n\right)$ and $\left(b_n\right)$ such that $\lim\left(a_n\right) = a = \lim\left(b_n\right)$; but $\lim\left(f\left(a_n\right)\right) \neq \lim\left(f\left(b_n\right)\right)$, then $\lim\limits_{x\to a} f(x) \nexists$.

Therefore, f is not continuous at a.

Example

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} \sin\frac{1}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}.$$

Prove that f is not continuous at $x = 0$.

Proof. We find a convergent sequence $\left(a_n\right)$ with $\lim\left(a_n\right) = 0$ but $\lim\left(f(a_n)\right) \neq f(0)$.

Let $a_n = \frac{1}{n\pi}$. Then $\lim\left(a_n\right) = 0$ but $\lim\left(f(a_n)\right) = 0 \neq 1 = f(0)$.

So, f is not sequential continuous at $x = 0$.

Hence, f is not continuous at $x = 0$.

Which theorem you prefer?

Alternative: Use Corollary 7.3.

Since $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$ does not exist, it follows that f is not continuous at $x = 0$.

Or by using definition of non continuity?

Examples

Evaluate (a) $\lim\left(m^{1/n}\right)$, where $m \in \mathbb{R}^+$ (b) $\lim\left(e^{1/n}\right)$

(c) $\lim\left(\sin\left(\frac{1}{n}\right)\right)$ (d) $\lim\left(\cos\left(\frac{1}{n}\right)\right)$ (e) $\lim\left(\tan\left(\frac{1}{n}\right)\right)$

Solution. (a)

(a) Let $p : \mathbb{R} \to \mathbb{R}$ be function defined by

$$p(x) = m^x, m \in \mathbb{R}^+ .$$

Then the function is continuous on $\mathbb{R}$.

Let $\left(x_n\right) = \left(\frac{1}{n}\right)$ be a sequence. Then

$$\lim\left(x_n\right) = 0 \Rightarrow \lim\left(m^{1/n}\right) = \lim\left(p\left(x_n\right)\right) = p(0) = 1 .$$

Solution. (b)

(b) Let $s : \mathbb{R} \to \mathbb{R}$ be function defined by $s(x) = e^x$.

Then the function is continuous on $\mathbb{R}$.

Let $\left(x_n\right) = \left(\frac{1}{n}\right)$ be a sequence. Then

$$\lim\left(x_n\right) = 0 \Rightarrow \lim\left(e^{1/n}\right) = \lim\left(s\left(x_n\right)\right) = s(0) = 1.$$

Solution. (c), (d), (e)

Let $f, g : \mathbb{R} \to \mathbb{R}$, $h : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$

be functions defined by

$f(x) = \sin x$, $g(x) = \cos x$ and $h(x) = \tan x$.

Then f and g are continuous for all $x \in \mathbb{R}$,

and h is continuous for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Let $\left(x_n\right) = \left(\frac{1}{n}\right)$ be a sequence. Then $\lim\left(x_n\right) = 0$ implies that

$$\lim\left(\sin\left(\tfrac{1}{n}\right)\right) = \lim\left(f\left(x_n\right)\right) = f(0) = 0\,.$$

$$\lim\left(\cos\left(\tfrac{1}{n}\right)\right) = \lim\left(g\left(x_n\right)\right) = g(0) = 1\,.$$

$$\lim\left(\tan\left(\tfrac{1}{n}\right)\right) = \lim\left(h\left(x_n\right)\right) = h(0) = 0\,.$$

Intermediate Value Theorem

Theorem 8.1. The Boundedness Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$.

Then f is bounded.

Proof. Question 12, Worksheet 5.

The theorem will be applied in the proof of Rolle's theorem in Chapter 5.

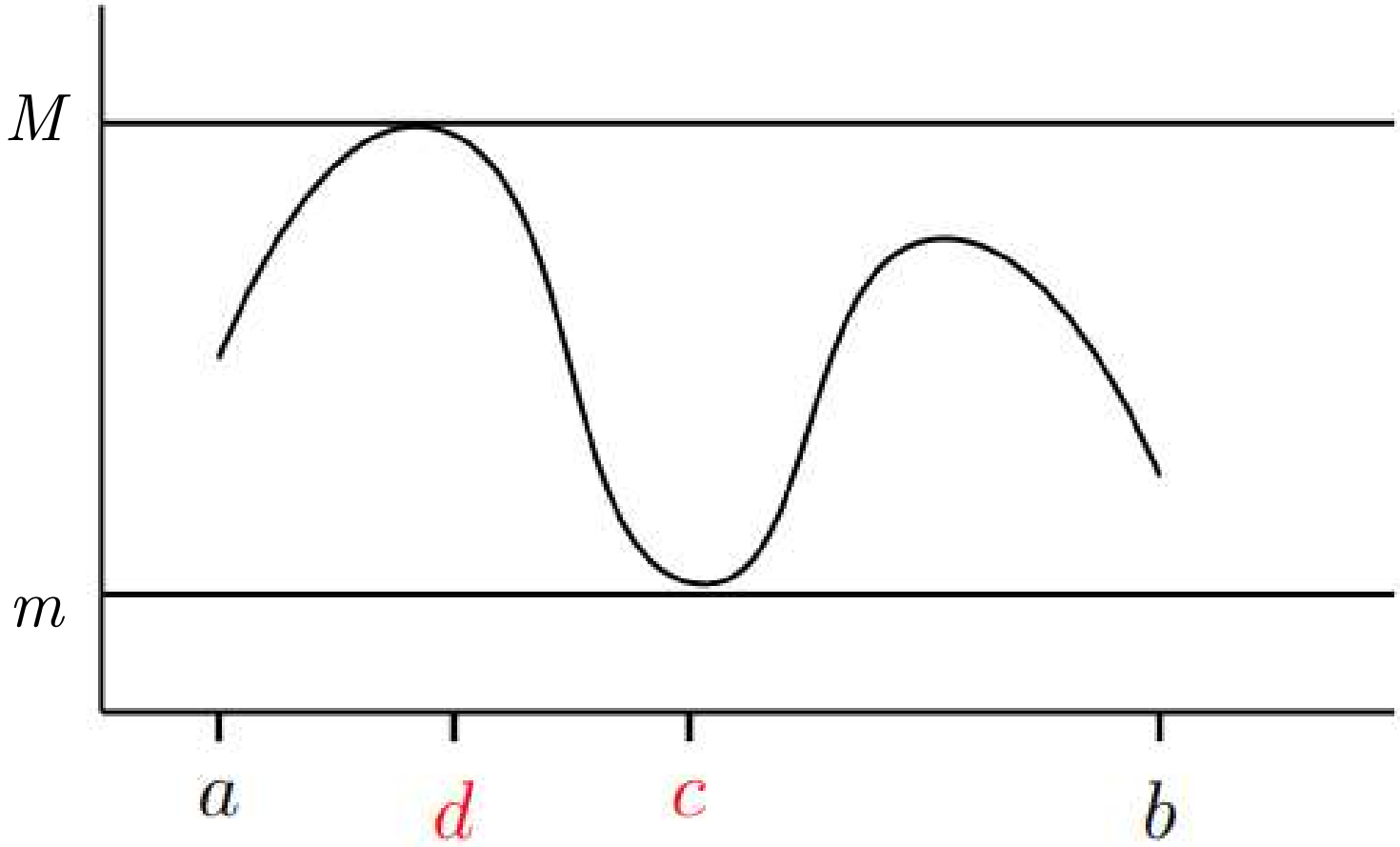
Theorem 8.2. Extreme Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$.

Then f attains its bounds on $[a, b]$.

Proof. Question 12, Worksheet 5.

The theorem will be applied in the proof of Rolle's theorem in Chapter 5.


M
m
a
d
c
b

Note that *Theorems 8.1 and 8.2* are not true on open intervals, e.g.,

$f(x) = \frac{1}{x}$ is continuous on $\left(0,1\right)$ but unbounded.

Question

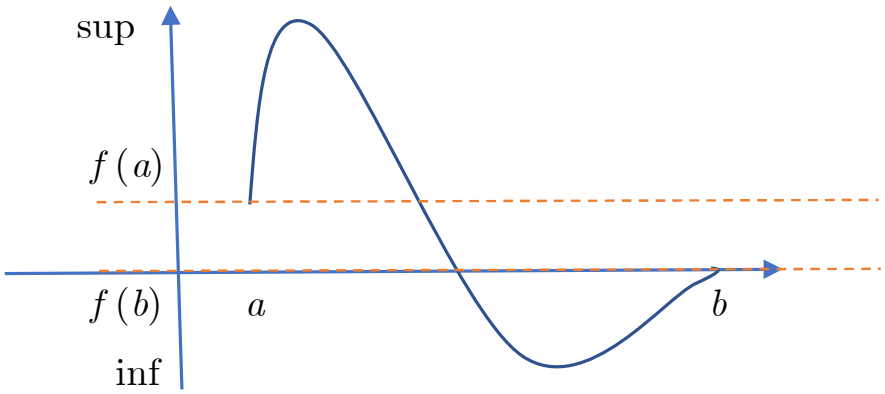
Is there exist some values of $x \in \Big(a, b\Big)$ such that

$f(a) < f(x) < f(b)$ (or $f(b) < f(x) < f(a)$)?

In English: Is there exist some values of $x \in \Big(a, b\Big)$ such that

$f(x)$ is in between $f(a)$ and $f(b)$?

$f(a)$ and $f(b)$ need not necessarily have to be the max/sup or min/inf.



Theorem 8.3. Intermediate Value Theorem

Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$.

Suppose $\alpha \in \mathbb{R}$ is in between $f(a)$ and $f(b)$

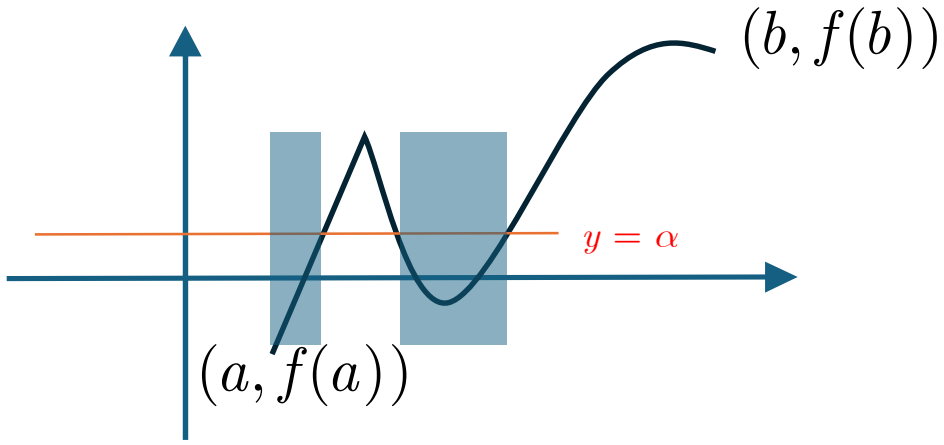
(i.e., $f(a) < \alpha < f(b)$ or $f(b) < \alpha < f(a)$).

Then $\exists c \in (a,b)$ such that $f(c) = \alpha$.

Proof. Without loss of generality, assume that $f(a) < f(b)$.

Let $\alpha \in \Big(f(a), f(b)\Big)$. Define $S = \Big\{x \in \Big[a, b\Big] : f(x) \leq \alpha\Big\}$.

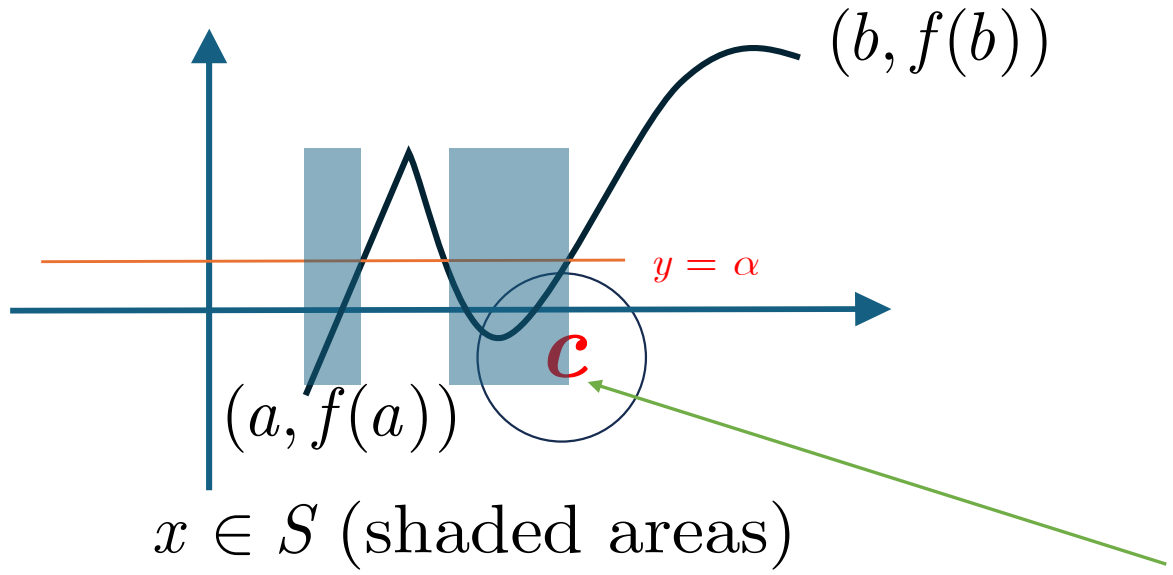
Note that $S \neq \varnothing$ ($\because a \in S$), a subset of $\mathbb{R}$ and bounded above.



$x \in S$ (shaded areas)

By the completeness axiom, $c = \sup S \in \mathbb{R}$ exists.

Furthermore, since $\left[a, b\right]$ is closed, $c \in S$, i.e., $c = \max S$.



We prove that $f(c) = \alpha$.

Since $c = \sup S$, it follows by Theorem 4.2(i) that

$\exists a_n \in S$ such that $c - \frac{1}{n} < a_n \leq c \ \forall n \in \mathbb{N}$.

Theorem 4.2(i) gives this part only, the other part is based on the definition of a_n.

How we construct the sequence (a_n)?

Theorem 4.2 (i). $c = \sup S \Leftrightarrow \forall \varepsilon > 0 \ \exists a \in S \ni c - \varepsilon < a$.

For all $\varepsilon > 0$, there exists an $a \in S$ such that $c - \varepsilon < a$.

When $\varepsilon = 1$, there exists an $a_1 \in S$ such that $c - 1 < a_1$.

When $\varepsilon = \frac{1}{2}$, there exists an $a_2 \in S$ such that $c - \frac{1}{2} < a_2$.

...

When $\varepsilon = \frac{1}{n}$, there exists an $a_n \in S$ such that $c - \frac{1}{n} < a_n$.

...

We construct a sequence $\left(a_n\right)$ by taking all the a_n above.

Note that a_n may not be distinct to a_m for any $n \neq m$

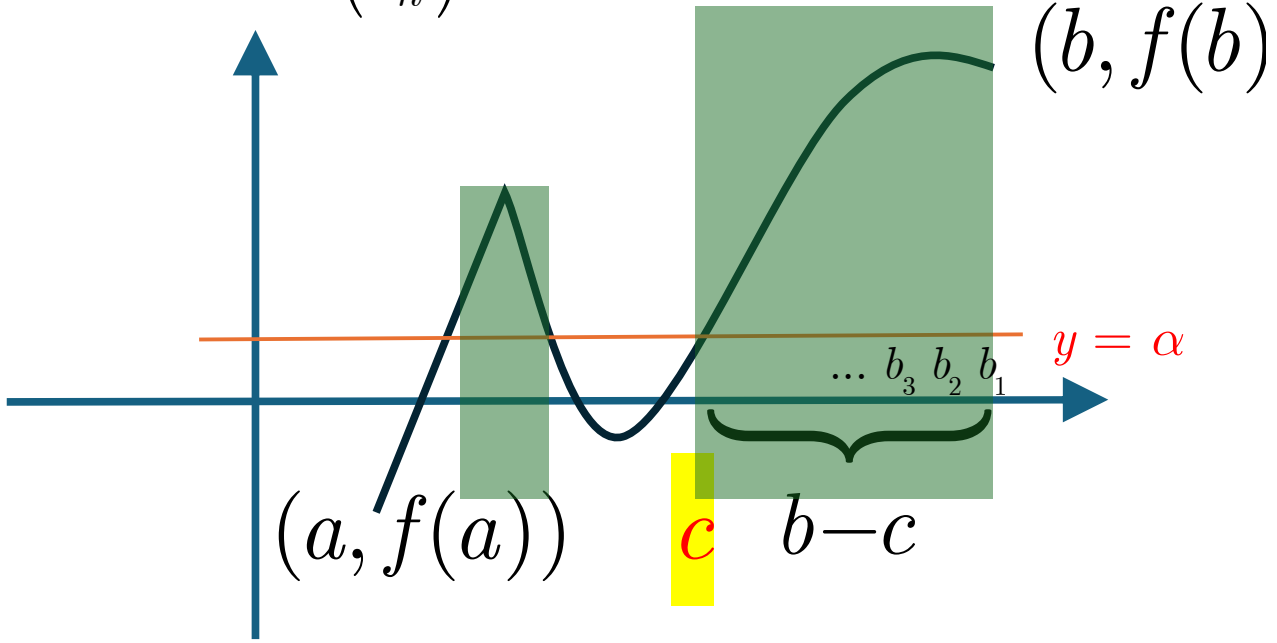
but that is fine because we are constructing a sequence.

Note that also $c - \frac{1}{n} < a_n \leq c \; \forall n \in \mathbb{N}$.

Now we let $T = \{x \in [a,b] : f(x) \geq \alpha\}$.

Define a sequence $b_n \in T$ by $b_n = c + \frac{b-c}{n}$.

Note that $\lim(b_n) = c$.



$x \in T$ (shaded areas)

It follows that $c - \frac{1}{n} < a_n \leq c \leq b_n = c + \frac{b-c}{n}$.

Apply the squeeze theorem (on $\left(a_n\right)$) and direct computation of $\lim\left(b_n\right)$, we get

$$\lim\left(a_n\right) = \lim\left(b_n\right) = c\,.$$

By Theorem 6.3 (Chapter 2),

$\lim\left(f\left(a_n\right)\right) \leq \alpha$ and $\lim\left(f\left(b_n\right)\right) \geq \alpha$

because $a_n \in S = \left\{x \in \left[a,b\right] : f(x) \leq \alpha\right\}$ and

$b_n \in T = \left\{x \in \left[a,b\right] : f(x) \geq \alpha\right\}$.

$\because f$ is continuous at c iff f is sequentially continuous at c.

$\therefore \lim\left(a_n\right) = \lim\left(b_n\right) = c \Rightarrow \lim\left(f\left(a_n\right)\right) = \lim\left(f\left(b_n\right)\right) = f(c)\,.$

From the previous slide, $\lim\left(f\left(a_n\right)\right) \leq \alpha$ and $\lim\left(f\left(b_n\right)\right) \geq \alpha\,.$

Hence, we conclude that $f(c) = \alpha\,.$

Next, we shall prove that $c \neq a, b$.

Prove by contradiction: Suppose $b = c$.

From previous results, we have

$\alpha \geq \lim\Big(f(a_n)\Big) = f(c) = f(b)$, a contradiction

since $\alpha \in \Big(f(a), f(b)\Big)$, i.e., $\alpha < f(b)$.

Now suppose $a = c$, then

$$f(a) = f(c) = \lim\Big(f(a_n)\Big) = \lim\Big(f(b_n)\Big) \geq \alpha\,,$$

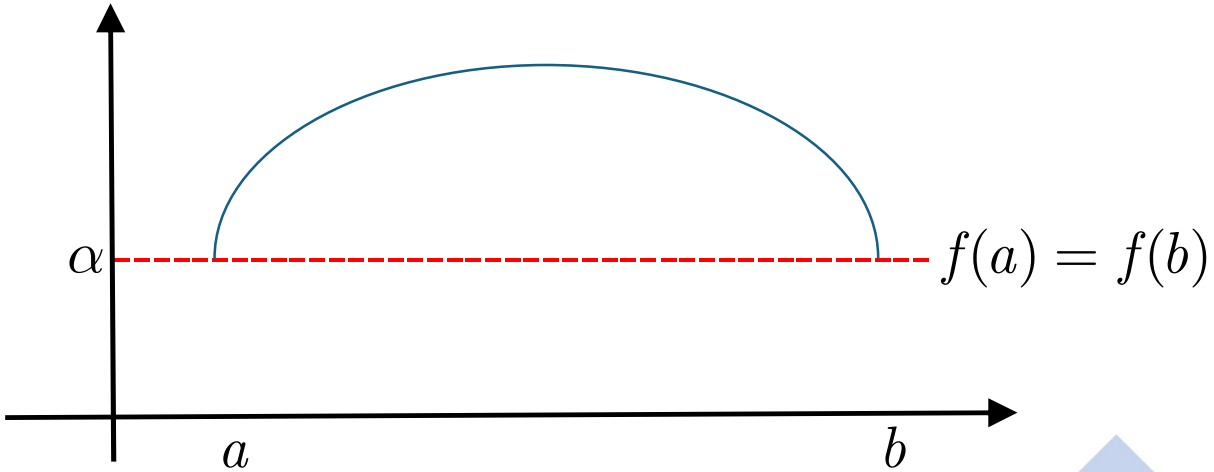
contradict to the fact that

$\alpha \in \Big(f(a), f(b)\Big)$, i.e., $\alpha > f(a)$.

Remark: Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$.

Suppose $\alpha \in \mathbb{R}$ such that $f(a) \leq \alpha \leq f(b)$ or $f(b) \leq \alpha \leq f(a)$.

Then there exists at least a $c \in [a, b]$ such that $f(c) = \alpha$.



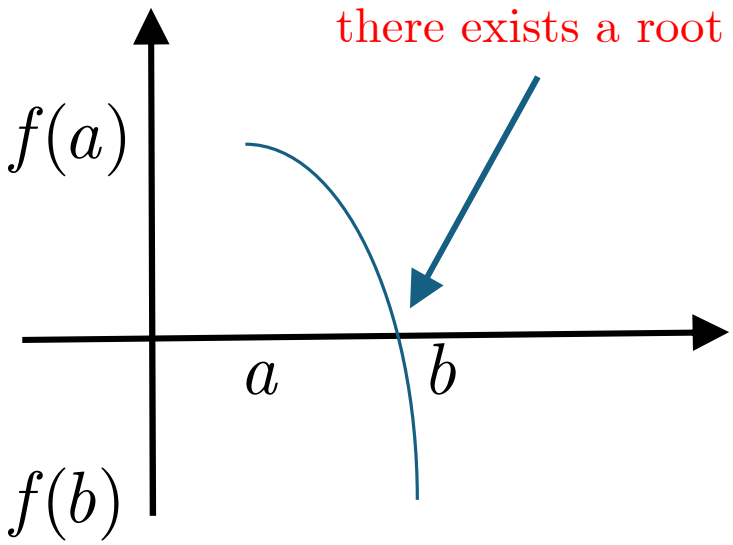
Corollary 8.4. Bolzano's theorem (Existence of roots). Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. If $f(a)f(b) < 0$, then there exists at least one root in between a and b.

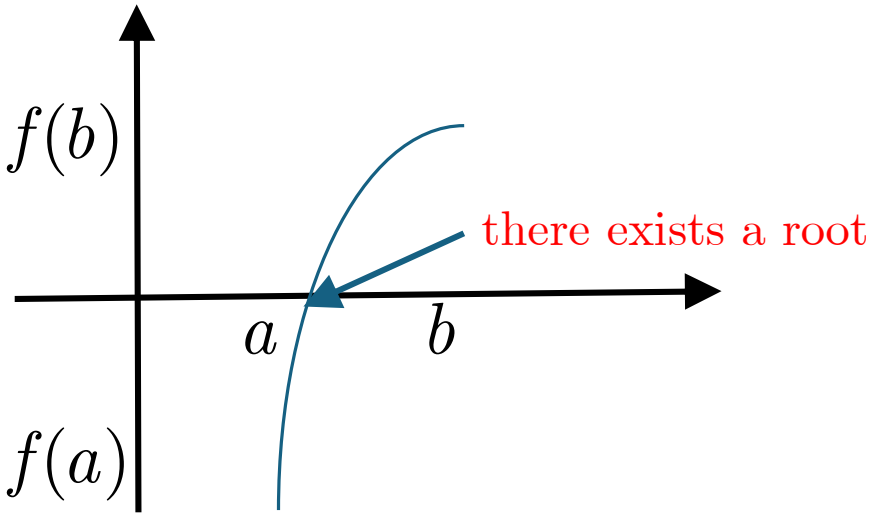
Proof. If $\alpha = 0$ is in between $f(a)$ and $f(b)$.

Then $f(a)$ and $f(b)$ must have different signs (one positive and one negative). So $f(a)f(b) < 0$.

By the IVT, $\exists c \in \left(a, b\right)$ such that $f(c) = \alpha = 0$,

i.e., f has (at least) a root in between $\left(a, b\right)$.


there exists a root
$f(a)$
$f(b)$
a
b


$f(b)$
there exists a root
a
b
$f(a)$

Example: Application of IVT (existence of roots)

Show that the following equation has a solution on the given interval.

(a) $e^x = 10x$ on $(0, 1)$.

(b) $17x^7 - 19x^5 = 1$ on $(-1, 0)$.

Solutions

Proof. (a) Let $f(x) := e^x - 10x$.

Then f is continuous on $\left[0,1\right]$ and

$f(0)f(1) = 1(e-10) < 0$.

Hence, by IVT, $\exists c \in \left(0,1\right)$ э $f(c) = 0$.

Proof (b)

(b) Let $g(x) := 17x^7 - 19x^5 - 1$.

Then g is continuous on $\Big[-1,0\Big]$ and

$g(0)g(-1) < 0 \because g(0) < 0, g(-1) > 0$.

Hence, by IVT, $\exists c \in \Big(-1,0\Big) \ni g(c) = 0$.

Example

Let $f:\left[0,2\right]\to\mathbb{R}$ be continuous on $\left[0,2\right]$ with $f(0)=f(2)$.

Prove that there exists an $c\in\left[0,1\right]$ with $f(c)=f(c+1)$.

Proof. Define $g:\left[0,1\right]\to\mathbb{R}$ by $g(x)=f(x+1)-f(x)$.

Then g is continuous on $\left[0,1\right]$ and

$g(0)=f(1)-f(0)$

$g(1)=f(2)-f(1)=f(0)-f(1)=-g(0)$

Proof.

Suppose $g(0) \neq 0$, then $g(0)g(1) < 0$.

By IVT, $g(c) = 0$ for some $c \in \big(0,1\big)$,

i.e., $f(c+1) = f(c)$.

Otherwise, if $g(0) = 0$.

Then $f(1) - f(0) = 0$ so $c = 0$ does the job.

Bisection method

The *bisection method* is an ancient root-finding method that applies IVT to any continuous function for which one knows two values with opposite signs, i.e., $f(a)f(b) < 0$, then there is a root in between a and b.

The method is to bisect a and b by taking their midpoint and determine either $f(a)f(\frac{a+b}{2}) < 0$ or $f(\frac{a+b}{2})f(b) < 0$.

Repeat the process finitely many times to get desired accuracy.

$f(x) := x^3 + x + 1 = 0\,.$

Iteration	Root is in between	Reason
1	$-1, 0$	$f(-1)f(0) < 0$
2	$-1, -\frac{1}{2}$	$f(-1)f(-\frac{1}{2}) < 0$ but $f(0)f(-\frac{1}{2}) > 0$
3	$-\frac{3}{4}, -\frac{1}{2}$	$f(-\frac{3}{4})f(-\frac{1}{2}) < 0$ but $f(-\frac{3}{4})f(-1) > 0$
4	$-\frac{3}{4}, -\frac{5}{8}$	$f(-\frac{3}{4})f(-\frac{5}{8}) < 0$ but $f(-\frac{5}{8})f(-\frac{1}{2}) > 0$
5	$-\frac{11}{16}, -\frac{5}{8}$	$f(-\frac{11}{16})f(-\frac{5}{8}) < 0$ but $f(-\frac{3}{4})f(-\frac{11}{16}) > 0$
6	$-\frac{11}{16}, -\frac{21}{32}$	$f(-\frac{11}{16})f(-\frac{21}{32}) < 0$ but $f(-\frac{21}{32})f(-\frac{5}{8}) > 0$
7	$-\frac{11}{16}, -\frac{43}{64}$	$f(-\frac{11}{16})f(-\frac{43}{64}) < 0$ but $f(-\frac{43}{64})f(-\frac{21}{32}) > 0$
8	$-\frac{11}{16}, -\frac{87}{128}$	$f(-\frac{11}{16})f(-\frac{87}{128}) < 0$ but $f(-\frac{87}{128})f(-\frac{43}{64}) > 0$
9	$-\frac{11}{16}, -\frac{175}{256}$	$f(-\frac{11}{16})f(-\frac{175}{256}) < 0$ but $f(-\frac{175}{256})f(-\frac{87}{128}) > 0$

Now $-\frac{11}{16} = -0.6875$ and $-\frac{175}{256} = -0.68359375\,.$

We know that the root is in between that two numbers. In fact,

By using calculator, the root is -0.6823 (correct to four decimal places).